

G.C.E. A/L Results Data Warehouse & OLAP Cube

Comprehensive Documentation of Quantitative Measures and Metrics

1. Executive Summary & System Context

This technical report outlines the quantitative measures and architectural design implemented for the G.C.E. Advanced Level results data warehouse. Built using DuckDB and Pandas inside Google Colab, the system processes 21+ million candidate records using a star schema to provide instant multi-dimensional analytical queries.

2. Quantitative Measures Used in the OLAP Cube

Measures represent the numeric performance metrics evaluated across multiple dimensions (Time, Academic Stream, Student Demographics, and Location). The OLAP_AL_Cube table incorporates four primary quantitative measures pre-computed via SQL aggregation:

1. Total Students (Additive Count Measure)

SQL Expression: `COUNT ( f.result_key )`

Description: Counts the total volume of student result entries falling within any specific slice, subtotal, or grand total grouping set (e.g., total candidates in the Arts stream for 2020).

2. Average Z-Score (Additive Average Measure)

SQL Expression: `AVG ( f.z_score )`

Description: Computes the mean academic performance score across student cohorts, enabling direct comparisons of student performance levels across years, streams, and genders.

3. Highest Z-Score (Maximum Metric Measure)

SQL Expression: `MAX ( f.z_score )`

Description: Identifies the maximum upper boundary of academic achievement within any dimensional slice.

4. Lowest Z-Score (Minimum Metric Measure)

SQL Expression: MIN(f.z_score)

Description: Identifies the minimum lower boundary of academic achievement within any dimensional slice.

3. Summary Matrix of Cube Measures

Measure Name	Data Type	Aggregation Function	Business Purpose & Analysis
total_students	INTEGER	COUNT()	Volume analysis, cohort sizing, and participation tracking
average_z_score	DOUBLE	AVG()	Central tendency evaluation of academic performance
highest_z_score	DOUBLE	MAX()	Peak achievement benchmarking by category
lowest_z_score	DOUBLE	MIN()	Baseline performance tracking

4. SQL Implementation for Cube Measures

The measures were compiled using grouping sets in DuckDB to pre-compute totals and subtotals across all dimensions:

```
CREATE OR REPLACE TABLE OLAP_AL_Cube AS
SELECT
  COALESCE(CAST(t.al_year AS VARCHAR), 'All Years') AS al_year,
  COALESCE(s.stream, 'All Streams') AS stream,
  COALESCE(st.gender, 'All Genders') AS gender,
  COUNT(f.result_key) AS total_students,
  AVG(f.z_score) AS average_z_score,
  MAX(f.z_score) AS highest_z_score,
  MIN(f.z_score) AS lowest_z_score
FROM Fact_AL_RESULTS f
JOIN Dim_Time t ON f.time_key = t.time_key
JOIN Dim_Stream s ON f.stream_key = s.stream_key
JOIN Dim_Student st ON f.student_key = st.student_key
WHERE f.z_score IS NOT NULL
GROUP BY GROUPING SETS (
  (t.al_year, s.stream, st.gender),
  (t.al_year, s.stream),
  (s.stream, st.gender),
  (t.al_year),
  (s.stream),
  (st.gender),
```

```
()  
);
```